

Semantic Audit: A Lean-Verified Single-Transition Theorem for CMV Ball Density

Independent semantic review completed and accepted by
openai-codex/gpt-6-astra

Verdict and scope

Equivalent to the frozen analytic profile claim. The primary declaration `CMVBallCase.typeC_dominance_persists` proves exactly that, for real parameters satisfying

$$0 < \lambda < 1, \quad \lambda\pi < v_0 < v, \quad I_C(v_0) \leq I_B(v_0),$$

one has

$$I_C(v) < I_B(v).$$

The weak initial inequality, strict later inequality, universal quantification, and strict area thresholds are unchanged. The profiles in this statement are the infima over all admissible angular candidates of the specified type. Neither prior strict dominance, existence of a crossing, nor a finite-perimeter classification is a hypothesis of the formal implication.

This review covers all 153 declarations in the exact publication inventory, their author-supplied statements and explanations, the relevant Lean proofs and uses, both assembled manuscripts, and the single external citation. The inventory reports 178 source declarations in the module closure but selects 153 for publication; the review list follows that exact selection. There are no configured generated families or indexed generated declarations.

Findings

- No critical mathematical mismatch was found in the primary claim, main proof, or supplement.
- All 153 declaration translations preserve the formal hypotheses, quantifier dependencies, domains, equality directions, and strictness. Definitions presented geometrically are explicitly distinguished from their total real-valued expressions.
- The body-only rules are satisfied by `main_markdown` and `supplement_introduction`. The main proof introduces the source, model, notation, citation, and exact result before the argument, and ends with qualified geometric scope and a conventional bibliography. The supplement introduction is heading-free prose. Renderer-owned front matter, section scaffolding, and HTML Lean-reference blocks are not author defects.
- The arXiv citation has an exact version and verified proposition, theorem, page, and figure locators. No unresolved citation issue remains.

Evidence and proof mechanism

The admissible angular domains are

$$0 < \beta \leq \pi - \arccos \lambda$$

for type B,

$$0 < b < \pi/2$$

for type C.

The inverse type B chart is the single formula $\pi/2 - \arctan((R - \lambda)/\sqrt{1 - \lambda^2})$, not the reciprocal acute-angle chart used outside its quadrant. Its inverse identities, admissibility, radius nonnegativity, and angular injectivity cover all $R \geq 0$, including $R < \lambda$. The type C inverse covers every positive radius. Radius-coordinate area and perimeter agree with the angular formulas on these domains.

The derivative identities give $A_{B'} = RP_{B'}$ and $A_{C'} = RP_{C'}$, with positive perimeter derivatives for positive radii. Continuity supplies the closed type B endpoint. The intermediate value arguments provide positive area radii for every $v > \lambda\pi$; strict area monotonicity and the inverse charts prove uniqueness among every admissible angular candidate. Thus the infimum definitions evaluate to actual candidate perimeters. Their use in the root does not exploit an empty set or totalized division.

The hard equal-radius inequality is established by the exact identity $P_B - P_C = 2H$. For $0 < R \leq \lambda$, direct term signs give $H > 0$. For $R > \lambda$, the proof differentiates the density ratio and uses

$$\partial_c(H(c, R)/c) = \frac{R}{c^2}L(c, R), \quad \partial_c L(c, R) = -\frac{cN(c, R)}{\sqrt{1 - c^2} D(c, R)^2}.$$

The polynomial N is strictly decreasing in density on $(-\infty, 1]$ for $R > 0$. The argument for $N(c, R) \geq 0$ includes the stationary locus $N(c, R) = 0$: N is strictly positive to its left, so L has strictly decreased from its continuous zero-density value. The $N < 0$ case uses $R > 1$ and the negative left limit $-b(R)$. The finite diagonal endpoint $c = R < 1$, the singular corner $R = 1$, and the region $R > 1$ are handled separately. At $R = 1$, a squeeze estimate controls the product $(1 - c)\beta_c(1)$; no spurious endpoint angle is assigned. Strict positivity from a zero limit uses an intermediate density, rather than mistaking nonnegativity for positivity. These steps establish the global equal-radius comparison, not an unproved monotonicity of the equal-area profile difference.

The common-perimeter inverses require $p > 2\lambda\pi$ as a subtype input and return positive radii with their perimeter equations. Existence and uniqueness are proved before choice is used. The equal-radius comparison forces $r_B(p) < r_C(p)$. Local inverse-function arguments then provide a representative of the common-perimeter area gap with

$$G'(p) = r_B(p) - r_C(p) < 0.$$

The formal clamping construction transports this gap to a real function only to apply interval calculus. Its derivative theorem is within $[p, \infty)$, including a one-sided derivative at the clamp point; it is not an ordinary derivative assertion across that point. The mean value theorem gives strict decrease of G . Weak profile dominance at v_0 yields $G(I_B(v_0)) \leq 0$, while strict increase of I_B yields $G(I_B(v)) < 0$ at every larger area. The final bridge recovers strict type C profile dominance. The main manuscript follows this actual proof mechanism.

Model and geometry fidelity

The density is λ on the closed unit disk and one outside. In particular, the type B unit-circle boundary arc has weight λ , not one. Its candidate encloses the unit disk, so the weighted area subtracts $(1 - \lambda)\pi$ from the two-segment Euclidean area. For type C, the deficit is $(1 - \lambda)$ times the overlap lens. With

$$S(t) = t - \text{sintcost},$$

the unit-radius segment of central angle $2t$ has area $S(t)$. The author's main and supplement explanations use this correct normalization. The nearby Lean source comment calling it twice a segment area does not describe the actual definition or proof: the exact expression, derivative $2\sin^2 t$, and reader-facing formulas have the correct scale.

The type B zero-radius endpoint is the unit disk and has area $\lambda\pi$ and perimeter $2\lambda\pi$. Its other angular endpoint is an infinite-radius limit and is excluded from finite candidates. Both type C endpoints are excluded; its zero-radius total expression is not used to prove existence. The root excludes $\lambda = 0, 1$, $v_0 = \lambda\pi$, and $v = v_0$, while explicitly allowing equality of the profiles at v_0 .

The formal objects are analytic real functions, not formalized finite-perimeter sets or weighted measures. Elementary chord, arc, and lens calculations connect these functions to the cited candidate families. Existence, regularity, connectedness, and exhaustiveness of geometric minimizers remain external. The manuscript correctly makes the conclusion $v_1 = v_2$ conditional on those source-level premises and the small-/large-area alternatives, rather than identifying it with the frozen root. Under those premises, continuous candidate profiles meet; persistence rules out a second meeting or later return of type B. This qualification is compatible with, and does not weaken, the requested analytic profile theorem.

Coverage and manuscript checks

Every inventory entry has exactly one review below, carrying its inventory `code_sha256` in the requested `source_sha256` field and its author's statement verbatim. Each reviewed reason records declaration-specific evidence rather than treating successful compilation as semantic equivalence. Generated-family coverage is empty because both the configured glob list and the inventory's generated assignments are empty; no sampled-family argument or numerical certificate is being substituted for a quantified proof.

A structural check compared both frozen Lean file hashes and the toolchain hash with the inventory/manifest, and recomputed every declaration excerpt hash. It also matched all 153 assembled supplement statements, explanations, source locators, hashes, and code excerpts to the reviewed handoff and source ranges. The assembled main body matches the author body after the expected Lean-token rendering, and its 110 Lean-reference tokens all resolve inside the inventory. The supplement distinguishes mathematical statements and explanations from source metadata and fenced code. The main mathematical displays and case splits were reviewed in the assembled Markdown; this review does not claim a new PDF rendering or a new Lean compilation.

External-source checks

The cited record is Cañete, Miranda Jr., and Vittone, *Some isoperimetric problems in planes with density*, arXiv:0906.1256v1 [math.DG], 8 June 2009. The versioned arXiv record was retrieved, and the bundled PDF identifies itself as that same version and date. In that PDF, **Proposition 3.22** begins on p. 32 and its classification proof continues on p. 33; type B encloses the unit disk and type C is an orthogonal ball. **Theorem 3.23** and the following conjectural equality-of-thresholds remark occur on p. 34, with the proof continuing through p. 36. The type B formulas are on p. 34, **Figure 19** and the type C formulas on p. 35. These are the exact locators used by the author. The formulas match the Lean definitions, including the weights and density deficits. The cited source is not represented as already proving the persistence theorem.

Disposition

Accept the `author-02` package as semantically equivalent to its frozen analytic claim. No critical author-body or supplement correction is required. This disposition rests on declaration-by-declaration comparison, the complete analytic argument and boundary analysis, manuscript consistency checks, and direct external-source verification. It is not an inference from compilation and is not certification of the unformalized global geometric-measure-theory premises.

Package-level findings

Relation: equivalent

No package-level semantic errors were reported.

Declaration-by-declaration audit

CMVBallCase.AdmissibleDensity

Source hash: 4e2cd6052fa41c3d2768e49b680c9466bd3d9e7beae3a567c7ebfa8e5256a410

Relation: equivalent

CMVBallDefinitions.lean:8-10: The definition is exactly $0 < \lambda$ and $\lambda < 1$, with no geometric or optimality premise. The author's open-interval description preserves both excluded constant-density endpoints.

Findings: none.

CMVBallCase.BAdmissible

Source hash: e044429fd7ffa85e0d7057933c7cab810b11565444bcff740b4688861a6b8c0d

Relation: equivalent

CMVBallCase.lean:18-22: The definition is precisely the conjunction $0 < \beta$ and $\beta \leq \pi - \arccos(\lambda)$. The explanation correctly separates this total predicate from the density hypothesis and retains the zero-radius endpoint.

Findings: none.

CMVBallCase.CAdmissible

Source hash: 0caecffaa369b9ef07b21d623e4e6c06f6a3df2ac3a26609e00159b4e2dad1f5

Relation: equivalent

CMVBallCase.lean:23-27: The definition is exactly $0 < \beta < \pi/2$. Both inequalities are strict; the explanation distinguishes limiting zero/infinite radii from totalized endpoint evaluations.

Findings: none.

CMVBallCase.CommonPerimeter

Source hash: e2043ced3f49efd166f38b27622f70ea021fb2c91e44ee48e02ae0fb32781041

Relation: equivalent

CMVBallCase.lean:1832-1840: The formal type is the subtype of real p satisfying $2 \cdot \lambda \cdot \pi < p$. The author's open interval description is exact for arbitrary λ ; positive-radius solvability and inverses are subsequently restricted to admissible density.

Findings: none.

CMVBallCase.arcSegment

Source hash: a0f37e78a16fab57786c18fa0ca15e0aec499e6edf77b3e0d77a98c1a7c7cdec

Relation: equivalent

CMVBallCase.lean:488-489: The definition is $x - \sin(x)\cos(x)$, exactly S . For a unit circle and angle $2x$ the sector-minus-signed-triangle calculation is S itself, not twice S ; the reader-facing explanation uses the correct normalization.

Findings: none.

CMVBallCase.arcSegment_hasDerivAt

Source hash: 32a91ac001b703b525fd39e1c6017868329abe912be28a6cd8b30291e39dfc4d

Relation: equivalent

CMVBallCase.lean:490-512: The source quantifies over arbitrary real f, f' , and x with HasDerivAt $f f' x$. Its conclusion is the chain rule coefficient $2 \cdot \sin(f(x))^2 \cdot f'$, exactly the author's statement, with no angle restriction.

Findings: none.

CMVBallCase.arcSegment_lt_self

Source hash: 7f8e1ec7190e9bd71f5255f6b427e61d471c4bb1d11fb61e4053360c07533a98

Relation: equivalent

CMVBallCase.lean:767-775: Lean assumes $0 < x < \pi/2$ and concludes the strict inequality $S(x) < x$. Positivity of both sine and cosine proves it; no non-strict endpoint claim is substituted.

Findings: none.

CMVBallCase.arcSegment_nonneg

Source hash: 984106fd2f39d9277591bc5e64a6ef86b070468729c98368039d0fdaf7f7e0f5

Relation: equivalent

CMVBallCase.lean:776-783: The formal domain is $[0, \pi]$. The proof treats 0 and π explicitly and invokes interior positivity elsewhere, matching the author's closed-interval statement.

Findings: none.

CMVBallCase.arcSegment_pos

Source hash: 4edb95bb9a7ef1f46aba008febfd569d56f56863cb28cc1fdf213082055ae376

Relation: equivalent

CMVBallCase.lean:610-622: The formal interval is $0 < x < \pi$. The proof splits on the sign of cosine and uses $\sin(x) < x$ in the positive-cosine case, giving strict positivity throughout the stated interval.

Findings: none.

CMVBallCase.arctan_neg_lam_div_sqrt_eq_arccos_sub_pi_div_two

Source hash: ae6498b57a7901ab66208434683619d4fcd6fd2d5539929922be1071e5b55472

Relation: equivalent

CMVBallCase.lean:163-182: The proof checks that $\arccos(\lambda) - \pi/2$ lies in the principal arctangent interval and computes its tangent. The translated identity has the correct sign, denominator, and density bounds.

Findings: none.

CMVBallCase.bAtRadius

Source hash: 4e3efb2ab3dd0b9f28df9260c5b5c743c4b28ba5b46ff3289c5cef1736b7d246

Relation: equivalent

CMVBallCase.lean:131-132: Lean defines $\arctan(1/R)$ as a total real expression. The author uses inverse-chart language only for $R > 0$ and correctly avoids attributing geometric meaning to the value at zero.

Findings: none.

CMVBallCase.bAtRadius_admissible

Source hash: f75ea7649633f8c47ad6d1362804bff9e3318e36e14147c0e5b256b30829c2cc

Relation: equivalent

CMVBallCase.lean: 284-289: The sole hypothesis $R > 0$ makes $1/R$ positive; arctangent is positive and strictly below $\pi/2$. The statement exactly identifies the open type C angular domain.

Findings: none.

CMVBallCase.bAtRadius_hasDerivAt

Source hash: 1653427eb62cf98e802e2428e9d6de180448223da78362c5f886faa270cd5aad

Relation: equivalent

CMVBallCase.lean: 393-402: The source requires $R \neq 0$, not positivity. The quotient and arctangent chain rules give $-1/(1 + R^2)$, and the author correctly includes negative nonzero inputs.

Findings: none.

CMVBallCase.bAtRadius_typeCRadius

Source hash: 79019d0015f2e75012a056920022fbe8538d02c8a466b649dea10f1677f1855a

Relation: equivalent

CMVBallCase.lean: 296-303: Every CAdmissible angle has a positive radius and is recovered by injectivity of the admissible angular radius map. The statement and proof concern all such angles, not a selected family member.

Findings: none.

CMVBallCase.betaAtRadius

Source hash: a3124126d7f948e4397701451937e291db6bb2593b860fca2145e77ec6602177

Relation: equivalent

CMVBallCase.lean: 127-130: The displayed arctangent formula agrees term for term with the total real definition. The explanation restricts its geometric inverse meaning to admissible density and nonnegative radii, without dropping the obtuse-angle branch.

Findings: none.

CMVBallCase.betaAtRadius_admissible

Source hash: 24ebcc1a37c0cdea1e90d38794d7abdbb4d018ce1729c36cafdffb499f7fb0029

Relation: equivalent

CMVBallCase.lean: 183-199: The source assumes $0 \leq R$, not $R > 0$, and proves both admissibility inequalities. Its comparison with the zero-radius arctangent gives the stated upper endpoint and includes $R = 0$.

Findings: none.

CMVBallCase.betaAtRadius_density_hasDerivAt

Source hash: 81dbbf3b42fcfbd5c1dad2e201bd766325c3f6f4bbc9066d7c22e79ab478c30f

Relation: equivalent

CMVBallCase.lean: 1437-1454: With R fixed but arbitrary, differentiating density gives $(1 - \lambda \cdot R)/(\sqrt{1 - \lambda^2} \cdot D)$. The author's formula, variable, and admissible-density hypotheses match the exact source coefficient.

Findings: none.

CMVBallCase.betaAtRadius_eq_arctan_reciprocal

Source hash: 1b10b001ba78b0255df00a8213bf477e9519c5b01dc94215f4dab6c80ce5263f

Relation: equivalent

CMVBallCase.lean: 1508-1514: The formal additional hypothesis $\lambda < R$ ensures a positive arctangent argument. The complementary arctangent identity then gives the stated reciprocal formula, with no illegitimate extension to the obtuse-angle branch.

Findings: none.

CMVBallCase.betaAtRadius_hasDerivAt

Source hash: 7bfaeb583b2974229111c194bad221441e7cff9af27728eb440c5e3349f59cc9

Relation: equivalent

CMVBallCase.lean: 378-392: The formal derivative is $-\sqrt{1 - \lambda^2} / ((R - \lambda)^2 + \sqrt{1 - \lambda^2}^2)$ for every real R under the two density bounds. With s as defined, the author's formula and fixed-density differentiation variable are exact.

Findings: none.

CMVBallCase.betaAtRadius_le_pi_div_two

Source hash: 6d70e241c44c4fc39707fea591fb93e2c57b4830b8617af8a63dd5d2033c3256

Relation: equivalent

CMVBallCase.lean: 148-156: The non-strict hypothesis $\lambda \leq R$ makes the arctangent argument nonnegative. The conclusion $\beta \leq \pi/2$ is likewise non-strict and includes $R = \lambda$, exactly as stated.

Findings: none.

CMVBallCase.betaAtRadius_pos

Source hash: 9bec3a5f14c974a03f3298e5c5d48fc6f60c6514fa386f2e0d0afaa8a38cf980

Relation: equivalent

CMVBallCase.lean: 133-138: Lean retains both open density hypotheses but places no condition on R . The upper bound $\arctan(x) < \pi/2$ proves the asserted positivity for every real radius expression.

Findings: none.

CMVBallCase.betaAtRadius_typeBRadius

Source hash: 2b33f238d62a297b2ebc523e21829901b4443f725ae563fae3e91de57dabbcb4

Relation: equivalent

CMVBallCase.lean:249-259: The statement quantifies over every admissible β . Nonnegativity of its radius, admissibility of its reconstructed angle, and radius injectivity prove the exact left-inverse identity in the explanation.

Findings: none.

CMVBallCase.betaAtRadius_zero

Source hash: 2b9379d0da8b78ab25eb062b33cc303712c06916a5c942063bd420dc473f31ab

Relation: equivalent

CMVBallCase.lean:272-277: Under $0 < \lambda < 1$ the inverse chart evaluates to $\pi - \arccos(\lambda)$. The proof uses the principal-angle identity rather than an arbitrary off-range inverse value.

Findings: none.

CMVBallCase.commonPerimeterAreaGap

Source hash: 71b55c7b076991056eaa30f6760e5f80aa24222bb5abfcede16ad3020d3377f0

Relation: equivalent

CMVBallCase.lean:2108-2116: The definition is A_B at the type B common-perimeter inverse minus A_C at the type C inverse. The author's sign, common-perimeter argument, and restricted domain are exact; this is not $I_B - I_C$ at equal area.

Findings: none.

CMVBallCase.commonPerimeterAreaGapAbove

Source hash: e2b63b71cf44b62491a0ebd0b8526e6a8332a7c063ef2fe57ee5f88f8d32ce4f

Relation: equivalent

CMVBallCase.lean:2235-2238: The definition composes G with `commonPerimeterClamp`. The author's real function $G_p(q) = G(\max(q, p))$ is exact and remains a composition using only genuine inverse arguments even below p .

Findings: none.

CMVBallCase.commonPerimeterAreaGapAbove_hasDerivWithinAt

Source hash: 2420c176e6ebf43330a3d88b7b4b75bdaec4c565775a067f362c214f4599e5c2

Relation: equivalent

CMVBallCase.lean:2239-2266: The formal derivative is within $\text{Ici } p.1$ at $q \geq p.1$, with coefficient $r_B(q) - r_C(q)$. The explanation faithfully states the one-sided endpoint interpretation and does not claim ordinary differentiability of the clamped function across $q = p$.

Findings: none.

CMVBallCase.commonPerimeterAreaGap_at_typeBProfile_nonpos

Source hash: 455cce8a30fd3fea9935e7a342738957b15a2a9f4b562858dc267332ef60912f

Relation: equivalent

CMVBallCase.lean:2334-2377: The formal premises are admissible density, $v > \lambda \cdot \pi$, and weak type C profile dominance. They imply the type C area radius is no larger than the type C common-perimeter radius, yielding $G(I_B(v)) \leq 0$ with exactly the sign stated.

Findings: none.

CMVBallCase.commonPerimeterAreaGap_local_hasDerivAt

Source hash: 8ac1ee8e9903b31ca5fa31187a2831b990514e5786275449b934fb9522a630a5

Relation: equivalent

CMVBallCase.lean:2117-2222: Lean quantifies for each admissible density and p first over a real representative gap, then gives its derivative at p and eventual agreement for every nearby admissible q . The author's neighborhood formulation preserves that quantifier order and does not assert a preselected global off-domain inverse.

Findings: none.

CMVBallCase.commonPerimeterAreaGap_strictAnti

Source hash: 91f23a2fb97afb962a3fafd5eb5a6db4289103e7ba32d905aa7120164cb1c340

Relation: equivalent

CMVBallCase.lean:2267-2320: StrictAnti is on the ordered CommonPerimeter subtype, equivalent to strict decrease on $(2\lambda \cdot \pi, \infty)$. The proof uses a continuous clamped representative and negative interior derivative on each $[p, q]$, exactly the explanation's mean-value argument.

Findings: none.

CMVBallCase.commonPerimeterClamp

Source hash: 4b4d169c7aed59d327b47ade1e0c9f86cfd6aa971b1e181dd947fb1ce217487c

Relation: equivalent

CMVBallCase.lean:2223-2226: The source value is $\max q \cdot p . 1$, with its domain proof from $\max q \cdot p . 1 \geq p . 1 > 2\lambda\pi$. The author states the same clamping operation for arbitrary real q without adding density bounds.

Findings: none.

CMVBallCase.commonPerimeterClamp_eq

Source hash: e3584ce3c3e58723599d9275ac8d07529840c51ccaf5b4e38a9fb3ce88756176

Relation: equivalent

CMVBallCase.lean:2227-2234: For $q \geq p . 1$, Lean proves equality of the clamped subtype with q equipped with its genuine-domain proof. The author's unchanged-value assertion correctly includes $q = p$ and preserves the absence of an admissible-density requirement.

Findings: none.

CMVBallCase.cos_bAtRadius_eq_radius_mul_sin

Source hash: 809c57b3c3abce1f1b8d70a94abd2bbd7102bca4c59fe3ebbe305dbaa59362d5

Relation: equivalent

CMVBallCase.lean:524-534: For $R > 0$, $\cot(b(R)) = R$ and positive $\sin(b(R))$ imply $\cos(b(R)) = R\sin(b(R))$. The author's equation and its use in the overlap-area derivative are exact.

Findings: none.

CMVBallCase.cot_betaAtRadius

Source hash: b6035ec4d3398844c66860e34566e7cf25d89bc622f0a05527f8328bc5c348c9

Relation: equivalent

CMVBallCase.lean:157-162: For the stated density bounds and arbitrary real R , complementary angles and $\tan(\arctan(x)) = x$ give the displayed quotient. No hidden $R \geq 0$ condition is introduced.

Findings: none.

CMVBallCase.cot_strictAntiOn_Ioo_zero_pi

Source hash: ace6f8c9441fb437f575beea814ae69da33538450d13d0301b36483ea0bd1558

Relation: equivalent

CMVBallCase.lean:92-102: The formal interval is the entire $(0, \pi)$, not just acute angles. Complementation transports its pairs to the principal tangent interval and reverses order, exactly as explained.

Findings: none.

CMVBallCase.existsUnique_typeBRadius_at_perimeter

Source hash: b6f0fa2f243c3e91136c9bd09e3537e769591131b17de77172ce5ad85508eb6b

Relation: equivalent

CMVBallCase.lean:1853-1884: Lean's existential uniqueness ranges over radii satisfying $R > 0$ and $P_B(R) = p$, with $p > 2\lambda \cdot \pi$ and admissible density. Its intermediate-value and strict-monotonicity proof matches the author's existence and uniqueness argument.

Findings: none.

CMVBallCase.existsUnique_typeCRadius_at_perimeter

Source hash: bd2558a1451eb56264500c5c5fcf17b65bf2502c53fb9be537238e05e61eadd9

Relation: equivalent

CMVBallCase.lean:1885-1922: The formal quantified condition is $R > 0$ and $P_C(R) = p$ on the genuine common-perimeter domain. The proof brackets the perimeter using $R = \lambda$ and a larger positive radius, so the author correctly avoids evaluation at $R = 0$.

Findings: none.

CMVBallCase.exists_typeBRadius_at_area

Source hash: 2eb4d55fd6e17591ebc46f3f94a245862a68631d4afc013b3f0eb27bc15ab111

Relation: equivalent

`CMVBallCase.lean`: 993-1019: Lean asserts existence, not independently uniqueness, of a positive radius for each $v > \lambda \cdot \pi$ under admissible density. The intermediate value proof starts at $A_B(0) = \lambda \cdot \pi$ and excludes zero by the strict area hypothesis, as stated.

Findings: none.

CMVBallCase.exists_typeCRadius_at_area

Source hash: 63e6ea9666dcdbe0cc4e7fdc659230e14fd24f25f1cdeb2756c507b3e9eb7e2f

Relation: equivalent

`CMVBallCase.lean`: 1020-1067: The formal existence domain is $v > \lambda \cdot \pi$ and the radius is strictly positive. The proof brackets v between the area at $R = \lambda$ and a large positive radius, avoiding the totalized $R = 0$ expression exactly as explained.

Findings: none.

CMVBallCase.fixedRadiusD

Source hash: 4e15dd87d6e310dd5beb90b522db2cdea3c545a1ce486ef9aaa593aa1c201858

Relation: equivalent

`CMVBallCase.lean`: 1245-1248: The source defines $(R - \lambda)^2 + (1 - \lambda^2)$. Expansion gives $R^2 - 2R \cdot \lambda + 1$, exactly the alternate expression in the explanation, with positivity asserted only later under density bounds.

Findings: none.

CMVBallCase.fixedRadiusD_density_hasDerivAt

Source hash: 8c046ed0856b99b8d7465ed1688893c4d60d5bff0186da511b6237c36d6dc073

Relation: equivalent

`CMVBallCase.lean`: 1455-1461: The formal derivative is $-2 \cdot R$ for all real λ, R . The expanded polynomial cancels both quadratic density terms, justifying the author's constant derivative without domain restrictions.

Findings: none.

CMVBallCase.fixedRadiusD_pos

Source hash: ab40250bb6c14fc69fd59cdf3f0cdb3baf9f85a08aed8a6aea9a5ea02fc5cf9a

Relation: equivalent

`CMVBallCase.lean`: 1364-1369: The two density hypotheses imply $1 - \lambda^2 > 0$, and the squared radius difference is nonnegative for every real R . The author's unrestricted-radius positivity claim exactly matches Lean.

Findings: none.

CMVBallCase.fixedRadiusHalfGap

Source hash: 7109dcf6ef8b06b1460afe81b7d4dc3cd06804fef3b67f56231578b2334248e2

Relation: equivalent

`CMVBallCase.lean`: 1240-1244: The definition agrees term by term with H , including the sign $\lambda - R$ and the coefficient $(1 - \lambda)R$ of b . Its total algebraic meaning and later restricted positivity claims are correctly separated.

Findings: none.

CMVBallCase.fixedRadiusHalfGap_density_ratio_continuousAt

Source hash: 63a18ea3889cd159ace5ef02e45f92fdd7d2a7fad2bca6f2d3d6f49921156755

Relation: equivalent

`CMVBallCase.lean`: 1660-1664: The source assumes $0 < c < 1$ and arbitrary real R and obtains continuity from the density derivative. The author preserves this unrestricted radius scope and correctly uses continuity at $c = R$ only when $0 < R < 1$.

Findings: none.

CMVBallCase.fixedRadiusHalfGap_density_ratio_hasDerivAt

Source hash: 8201756840d20049cb38f7556cdc3b5c84839cfb60b6a731c3c4da0399a6d421

Relation: equivalent

`CMVBallCase.lean`: 1480-1500: The source proves the ordinary density derivative $(R/\lambda^2) \cdot L$ for every real R under the two density bounds. The author correctly separates this identity from the additional $R > \lambda$ condition needed for negativity.

Findings: none.

CMVBallCase.fixedRadiusHalfGap_density_ratio_strictAntiOn

Source hash: 2f0edd3e2968d32be31b213f6dbe635d46d10fd277b8e4e89a8b9f3fd3a0d39e

Relation: equivalent

`CMVBallCase.lean`: 1640-1654: For fixed $R > 0$ the formal density interval is exactly $(0, \min(1, R))$. The factor R/c^2 is positive and L negative there; the author neither includes density zero nor extends beyond $c < R$.

Findings: none.

CMVBallCase.fixedRadiusHalfGap_density_ratio_tendsto_one

Source hash: 38b39009304f9c27c735560f1daa9451bc247788bdacf9b12a22ee47421536b1

Relation: equivalent

`CMVBallCase.lean`: 1726-1733: The source combines $R = 1$ and $R > 1$ to prove the pointwise left limit for all $R \geq 1$. The explanation correctly disclaims any uniformity in R .

Findings: none.

CMVBallCase.fixedRadiusHalfGap_density_ratio_tendsto_one_at_one

Source hash: 72d28fc306e34fd07f9a091bc1f04b99fa14efc650ed4c7075d01cf65333b270

Relation: equivalent

CMVBallCase.lean: 1695-1725: There are no free density or radius hypotheses: R is exactly one and density tends to one from below. The source squeezes $(1 - c)\beta_c(1)$ rather than evaluating the singular chart, exactly the boundary argument given by the author.

Findings: none.

CMVBallCase.fixedRadiusHalfGap_density_ratio_tendsto_one_of_gt

Source hash: ec5c31b64505c09d08c58f3f1b7efe55ef00bdd5aad235b9e99482e73cfe7914

Relation: equivalent

CMVBallCase.lean: 1665-1694: The formal statement is a left density limit to zero for each fixed $R > 1$. The proof replaces the chart by its reciprocal near one and takes the quotient limit with nonzero limiting denominator, as explained.

Findings: none.

CMVBallCase.fixedRadiusHalfGap_pos_of_gt

Source hash: fe2e7d9f4b2c9b099c9f8b560428cb8beb3d38b8a8d99f10c67f26ccbe96aa81

Relation: equivalent

CMVBallCase.lean: 1773-1782: The formal hypotheses are admissible density and $R > \lambda$. The proof splits $R < 1$ versus $R \geq 1$ with no missing equality case, exactly the author's complete high-radius region.

Findings: none.

CMVBallCase.fixedRadiusHalfGap_pos_of_le

Source hash: 70db5ffdb68d856e4b264b6a6571be92f1d5430f1f89edab30d115fd2ca0a8

Relation: equivalent

CMVBallCase.lean: 1386-1399: The formal assumptions are $0 < \lambda < 1$ and $0 < R \leq \lambda$. The first term of H is nonnegative and the next two are positive, including the equality case $R = \lambda$, exactly as stated.

Findings: none.

CMVBallCase.fixedRadiusHalfGap_pos_of_lt_radius_lt_one

Source hash: fbe9df98337f6aceb7cf9c5bfeb2a39837777085850f68cc2b9e8dfc133753b5

Relation: equivalent

CMVBallCase.lean: 1734-1753: Lean assumes $0 < \lambda < R < 1$, proves strict decrease on the half-closed interval $(0, R]$, and uses the positive diagonal ratio. The author's strict comparison with $c = R$ is thus justified, not inferred from continuity alone.

Findings: none.

CMVBallCase.fixedRadiusHalfGap_pos_of_one_le_radius

Source hash: e873740cbb342f72d438af00f9b3a7382b645a81e5e013505d7f0ed284853138

Relation: equivalent

CMVBallCase.lean:1754-1772: Under admissible density and $R \geq 1$, the proof first obtains nonnegativity at a larger interior density from the endpoint limit, then uses strict decrease to get strict positivity at λ . The author preserves this essential intermediate-point step.

Findings: none.

CMVBallCase.fixedRadiusL

Source hash: f9ee2e168349ec1a77b121c311033b4f50ae517f2839c175c84616e6784fbea2

Relation: equivalent

CMVBallCase.lean:1249-1253: The formal expression is $\text{betaAtRadius} - \text{bAtRadius} - \lambda \sqrt{1 - \lambda^2} / D$. The author has the correct minus sign and explicitly limits the negative-sign theorem to $R > \lambda$ rather than all positive radii.

Findings: none.

CMVBallCase.fixedRadiusL_continuousAt_zero

Source hash: d2266e6ab3843cad1067d095c347fbf2258071549d62313f18f286686d040702

Relation: equivalent

CMVBallCase.lean:1515-1535: Lean assumes $R > 0$ and proves ordinary continuity in density at zero. The proof checks the two denominators there are nonzero; the author's explanation identifies them correctly and does not claim a derivative at zero.

Findings: none.

CMVBallCase.fixedRadiusL_density_hasDerivAt

Source hash: db1661fbfdaeefec8e9d4f2e678214b5a54356b68cb7f3953f8533e10e578e7d

Relation: equivalent

CMVBallCase.lean:1462-1479: Lean proves the density derivative $-\lambda \cdot N / (\sqrt{1 - \lambda^2} \cdot D^2)$ for arbitrary R under $0 < \lambda < 1$. Its proof uses the positive denominator and square-root-square identity, and the explanation preserves the exact coefficient and scope.

Findings: none.

CMVBallCase.fixedRadiusL_neg

Source hash: 7412414d7e4f71205db23e4c7a9ba2c16e8de6f3ee9f630908e2c9dd59858948

Relation: equivalent

CMVBallCase.lean:1630-1639: The source requires $\lambda < R$ in addition to the two density bounds. Its complete split at $R = 1$ and at $N = 0$ proves strict negativity only on that stated domain, as the author expressly emphasizes.

Findings: none.

CMVBallCase.fixedRadiusL_neg_of_N_neg

Source hash: 7e62014feed7939abad45447e00b04f292d440a21f48f2f7d10485082ddd4696

Relation: equivalent

CMVBallCase.lean:1604-1629: The formal assumptions are $0 < \lambda < 1$, $R > 1$, and $N(\lambda, R) < 0$. The proof makes L increase to its negative left limit, yielding the author's bound $L(\lambda, R) \leq -b(R) < 0$ with correct direction.

Findings: none.

CMVBallCase.fixedRadiusL_neg_of_N_nonneg

Source hash: e7586227d8c6bfcd2ebb1b0a33219042c5b328676e83c2a41deef131c44f22f

Relation: equivalent

CMVBallCase.lean:1571-1594: All formal hypotheses, including $\lambda < R$ and $N(\lambda, R) \geq 0$, appear in the translation. The proof uses $N(c, R) > 0$ strictly to the left and continuity at zero, so its strict conclusion remains valid when $N(\lambda, R) = 0$.

Findings: none.

CMVBallCase.fixedRadiusL_tendsto_one

Source hash: 13ea5fd875867d8d70d1cab8b12f6ea2839f979405c944fc74ef550ef86c0035

Relation: equivalent

CMVBallCase.lean:1536-1570: The source limit is along $\text{nhdsWithin } 1 \text{ (Iio } 1)$, with the strict hypothesis $R > 1$. The reciprocal chart and $D(1, R) = (R - 1)^2$ give $-b(R)$; the author correctly excludes $R = 1$ and does not substitute the totalized value at density one.

Findings: none.

CMVBallCase.fixedRadiusL_zero

Source hash: 29c562fbf4f67e14b5c469f6f6f6dcfe57423601de3464b20c18a1c168e35d8e

Relation: equivalent

CMVBallCase.lean:1501-1507: For $R > 0$ the positive reciprocal arctangent identity gives $\beta_0(R) = b(R)$, while the correction term vanishes. The source and author both assert exactly $L(0, R) = 0$ on this positive-radius domain.

Findings: none.

CMVBallCase.fixedRadiusN

Source hash: c72277ca17b4cfcd6ad9cc4b941bef23e2a9af37be01cd916422a0b4b63e81c1

Relation: equivalent

CMVBallCase.lean:1361-1363: The polynomial coefficients, density variable, and radius variable match Lean exactly. The explanation correctly relates it to $-\lambda \cdot N/(\sqrt{1 - \lambda^2} \cdot D^2)$, not to N with the opposite sign.

Findings: none.

CMVBallCase.fixedRadiusN_density_derivative_identity

Source hash: 80b7c6ee3bbac239cfec59bac30eb2868a417e75a97013642551d21b2c7f23ee

Relation: equivalent

`CMVBallCase.lean`: 1370-1379: Lean proves `HasDerivAt` in the density variable for arbitrary real λ and R . Differentiating the polynomial gives $-4R^2 + 4R \cdot \lambda - 2$, equal to the completed-square expression displayed by the author.

Findings: none.

CMVBallCase.fixedRadiusN_density_derivative_neg

Source hash: 0a45acd25d263bfc0ab66ae247903b4ac172086632c9ab50764b1204d06b8751

Relation: equivalent

`CMVBallCase.lean`: 1407-1412: The source assumes $\lambda < 1$ and $R > 0$, with no lower bound on λ . Its completed-square coefficient is strictly negative because of the positive constant and nonnegative square; the author preserves this wider density domain.

Findings: none.

CMVBallCase.fixedRadiusN_diagonal

Source hash: 1e28106ded58409dc99d7ec87ff622e35ba249710e63bf8dc7c272196b448794

Relation: equivalent

`CMVBallCase.lean`: 1413-1417: Substituting $\lambda = R$ gives $R - R^3 = R(1 - R^2)$ for every real R . The nonnegative comparison at $0 < R \leq 1$ is a stated use, not an additional hypothesis of the identity.

Findings: none.

CMVBallCase.fixedRadiusN_pos_of_radius_le_one

Source hash: 02c15b97515cc50c1a2b220426d81b6aedc3b955c8b35761b70257e60ada615f

Relation: equivalent

`CMVBallCase.lean`: 1595-1603: The source assumes $0 < \lambda < R \leq 1$; $\lambda < 1$ follows and is not separately required. Strict decrease of N compared with the nonnegative diagonal value gives the stated strict positivity, including $R = 1$.

Findings: none.

CMVBallCase.fixedRadiusN_strictAntiOn

Source hash: 25b0b9b9ea43f5e32fc07e9659bfab1ecf97131d024c9759e860808c59663501

Relation: equivalent

`CMVBallCase.lean`: 1418-1427: For $R > 0$ Lean proves strict decrease on `Iic 1`, including density one and all negative densities. Continuity of the polynomial and strict interior derivative negativity justify exactly the author's domain.

Findings: none.

CMVBallCase.perimeter_gap_eq_two_mul_halfGap

Source hash: 8b62e7839212847fab2541e10be82a3d8159e4fde430a8284b2a3ed1383c3a75

Relation: equivalent

CMVBallCase.lean:1380-1385: The source is an unconditional identity for all real λ and R , proved by expansion and ring algebra. The author preserves the orientation $P_B - P_C$ and the factor 2.

Findings: none.

CMVBallCase.sameRadius_typeB_gt_typeC

Source hash: 9c6c3244ad6f03f01ef9a5860148ac78a1967d0c98c4165b638dc559db07fdf5

Relation: equivalent

CMVBallCase.lean:1783-1799: Lean asserts $P_C(R) < P_B(R)$ for all admissible densities and every $R > 0$. The proof combines $R \leq \lambda$ with $R > \lambda$ via the half-gap identity; the author correctly identifies this as equal-radius, not equal-area, comparison.

Findings: none.

CMVBallCase.sameRadius_typeB_gt_typeC_of_le

Source hash: 5dd1341e2c387a274bfb506d761e3f1b5f27cc70f720d3f55540418d8785332e

Relation: equivalent

CMVBallCase.lean:1400-1406: Under the same low-radius hypotheses, positivity of H and the identity $P_B - P_C = 2H$ give $P_C < P_B$. The author correctly presents this as a restricted equal-radius comparison.

Findings: none.

CMVBallCase.sin_bAtRadius_mul_cos

Source hash: 10fbbcdf677fa47ab4e65fa4fcec664bfcee9cb8b5e893c8c90bb4f234de0fd3

Relation: equivalent

CMVBallCase.lean:424-440: The formal hypothesis is only $R \neq 0$. The arctangent product simplifies to $R/(1 + R^2)$, with the correct sign at negative R , as the explanation states.

Findings: none.

CMVBallCase.sin_betaAtRadius_mul_cos

Source hash: e81216d3d0ee48ad5f9aec6192ac30e08e48ebe31d1c00ea3a7ac17be472a58d

Relation: equivalent

CMVBallCase.lean:403-423: For admissible density and arbitrary R , complementary-angle identities and the arctangent sine/cosine formulas give $(R - \lambda) \cdot s / ((R - \lambda)^2 + s^2)$. The author preserves the possibly negative numerator and unrestricted radius domain.

Findings: none.

CMVBallCase.sin_typeBAlpha_betaAtRadius

Source hash: 6ddcb65dd46c70fb4136025d8ae05d542b93e1cc2d2cac116509821e0248fdb0

Relation: equivalent

CMVBallCase.lean:513-523: Under admissible density and $R \geq 0$, multiplying the radius equation by the nonzero $\sin(\beta)$ gives $\sin(\beta + a) = R\sin(\beta)$. This is the cancellation used in the type B area derivative.

Findings: none.

CMVBallCase.sq_sqrt_one_sub_sq

Source hash: 398e78a14ca43373831b820093b0241c39626cd64fae57944b26894a61e0287e

Relation: equivalent

CMVBallCase.lean:58-62: Under the stated open density bounds, `Real.sq_sqrt` identifies the squared square root with $1 - \lambda^2$. No sign choice or radius hypothesis occurs.

Findings: none.

CMVBallCase.sqrt_one_sub_sq_hasDerivAt

Source hash: 1f9fdbbe32f5f9bb2811002baeba3f0bd3fa1478086f09a6412852133afef60cf

Relation: equivalent

CMVBallCase.lean:1428-1436: At $0 < \lambda < 1$ the positive radicand permits square-root differentiation, with coefficient $-\lambda/\sqrt{1-\lambda^2}$. The source and explanation differentiate density, not radius, and impose the same bounds.

Findings: none.

CMVBallCase.sqrt_one_sub_sq_pos

Source hash: 0bd6d03a5431ddb7e353eee713defc6c88eb6f034d3e70eb881f76b26abc6488

Relation: equivalent

CMVBallCase.lean:53-57: The two Lean hypotheses are exactly $0 < \lambda$ and $\lambda < 1$. They imply a strictly positive radicand, and `Real.sqrt_pos` supplies the asserted positive square root used in cancellations.

Findings: none.

CMVBallCase.typeBAlpha

Source hash: b162ed041fbf8301c033d26810d7ba715023e6c12a98223c2a0238ca585853e2

Relation: equivalent

CMVBallDefinitions.lean:11-13: Lean defines $\beta + \arccos(\lambda)$ for all real inputs. This is precisely the author's α , with its half-angle interpretation and $0 < \alpha \leq \pi$ restricted in the explanation to the admissible geometric domain.

Findings: none.

CMVBallCase.typeBAlpha_sin_nonneg

Source hash: 84b0b6514b1ce040ec07e3234a2e8730ab3f09c5e1a14930d078033a2d70a73f

Relation: equivalent

CMVBallCase.lean:72-80: Lean has only BAdmissible λ β , without density bounds. The proof uses global nonnegativity of $\arccos$ to put $\beta + \arccos(\lambda)$ in $[0, \pi]$; the author explicitly preserves this wider density domain and non-strict conclusion.

Findings: none.

CMVBallCase.typeBArea

Source hash: c81a1639e7c0ba21669318475c1f246edae8a287721cbc51038eb28187feccf8

Relation: equivalent

CMVBallDefinitions.lean:22-29: The source expression and the author's displayed formula agree term by term. Using $S(\pi - \beta) = \pi - \beta + \sin(\beta)\cos(\beta)$, the two segment areas and full-unit-disk density deficit reproduce the cited CMV formula.

Findings: none.

CMVBallCase.typeBAreaAtRadius

Source hash: 302ee62cf6614993146d2f898718e7b4c35270995d17efc0e3fbe06584054ba3

Relation: equivalent

CMVBallCase.lean:326-334: Expanding S gives precisely the two trigonometric segment terms and $-(1 - \lambda) \cdot \pi$ in Lean. The density correction is for the entire unit disk, consistent with the source's type B geometry.

Findings: none.

CMVBallCase.typeBAreaAtRadius_agrees

Source hash: c9cf34f1eb5dafbc32c5b2d81a1f3960d3e2a3fa29468cbef2fa5572ed5aa637

Relation: equivalent

CMVBallCase.lean:352-357: The exact inverse-radius identity rewrites the angular area into the radius expression with the same density correction. The author's $R \geq 0$ domain and density hypotheses match the source.

Findings: none.

CMVBallCase.typeBAreaAtRadius_continuous

Source hash: 654d93958fa1cc148933f3dbe93ac2d4925ccbef30b42f7a8afd01edc8ef532d

Relation: equivalent

CMVBallCase.lean:693-697: Lean has no density condition and concludes Continuous on all real radii. Division by a fixed real constant is continuous even when totalized at zero; the author explicitly explains this wider algebraic scope.

Findings: none.

CMVBallCase.typeBAreaAtRadius_deriv_pos

Source hash: 4fcb16a2eb66bac06f3aac56a18581ba3fae227c21397173be1ab0d712a98430

Relation: equivalent

CMVBallCase.lean:663-669: The formal conclusion is positivity of $\text{deriv}A_B$ at positive R . The proof rewrites the established derivative and uses $R > 0$ and $Q_B > 0$, matching the explanation without asserting positivity at zero.

Findings: none.

CMVBallCase.typeBAreaAtRadius_hasDerivAt

Source hash: e5d44f049460bf59353073b4b5ae0475e767191b44bc8056fa589b30ce308ada

Relation: equivalent

CMVBallCase.lean:535-571: The formal hypotheses are $0 < \lambda < 1$ and $R > 0$. The proof cancels the two angle-derivative contributions using $\sin(\beta + a) = R\sin(\beta)$, leaving $R \cdot (2Q_B)$, equivalent to $2RQ_B$ and RP_B' .

Findings: none.

CMVBallCase.typeBAreaAtRadius_large_lower

Source hash: 141341249a38db9c1c4d40b1c1642f4101a23d6266b432cbe39ad76a061df4b3

Relation: equivalent

CMVBallCase.lean:784-809: The source assumes $\lambda \leq R$ and $0 < \lambda < 1$ and concludes $R^2 \cdot (\pi/2) - \pi < A_B(R)$. The author preserves strictness and the lower radius restriction; nonnegative segment terms and a strictly smaller-than- π density correction justify the bound.

Findings: none.

CMVBallCase.typeBAreaAtRadius_strictMonoOn

Source hash: ab4db7494426ac1cdccec00e6e10d30da1a1b7a3f1a760b845607901df77bbd3

Relation: equivalent

CMVBallCase.lean:713-721: The formal domain is the closed half-line $\mathbb{I}c\mathbb{i} \ \emptyset$ under $0 < \lambda < 1$. Continuity there and positive derivative on its interior justify the mean-value argument, including comparisons with zero.

Findings: none.

CMVBallCase.typeBAreaAtRadius_zero

Source hash: 6d532317078066c3097d5d0410644fac9c192ee7d6af6804f7e2c0d6a360b53d

Relation: equivalent

CMVBallCase.lean:749-758: The proof reduces the angular endpoint to $\alpha = \pi$ and radius zero, giving $\pi - (1 - \lambda)\pi = \lambda \cdot \pi$. The statement preserves admissible density and the exact endpoint area.

Findings: none.

CMVBallCase.typeBPerimeter

Source hash: 3f47b16460dd98ef1ba0cd3846453bf4afaacfb39faf78478473722bd9ce0388

Relation: equivalent

CMVBallDefinitions.lean:18-21: The formal total expression is $2 \cdot ((\pi - \beta) \cdot r + \lambda \cdot \alpha)$. The source's exterior and unit-circle arc lengths yield exactly these weights, including density λ on the closed unit-circle boundary.

Findings: none.

CMVBallCase.typeBPerimeterAtArea

Source hash: 98f0853f0439181bac4c27175769bb69c6bb2e98d9167f9ffa0f7bf07436ebdd

Relation: equivalent

CMVBallCase.lean:1080-1084: The definition is sInf of the all-candidate type B perimeter set for arbitrary real inputs. The explanation accurately distinguishes the total real infimum from its later nonempty-singleton evaluation on the root's area domain.

Findings: none.

CMVBallCase.typeBPerimeterAtArea_eq

Source hash: 57097d90269686a2eff890ae3d5ae1d3b35601393a261dbde597136f7ab0b41c

Relation: equivalent

CMVBallCase.lean:1124-1130: The statement has the candidate's admissibility, area equality, and universal angular uniqueness as hypotheses. The proof applies the singleton-set lemma and csInf_singleton , justifying the author's conditional profile evaluation.

Findings: none.

CMVBallCase.typeBPerimeterAtArea_eq_radius

Source hash: 28e93575ceb4de605bf1e8801ce418243463ddd89e855c02916af6db2f66c59d

Relation: equivalent

CMVBallCase.lean:1165-1182: Lean assumes positive R with $A_B(R) = v$ and admissible density, but no separate $v > \lambda \cdot \pi$ hypothesis. The proof establishes uniqueness of every admissible angular candidate through area and radius injectivity, exactly as explained.

Findings: none.

CMVBallCase.typeBPerimeterAtArea_gt_commonEndpoint

Source hash: 1bc2f925dd32fdc3436ba53a34596445576e91faa97e786b1962cb23ce1fbf2b

Relation: equivalent

CMVBallCase.lean:2321-2333: For $v > \lambda \cdot \pi$ and admissible density, a positive area radius and strict perimeter increase from $R = 0$ prove $2\lambda \cdot \pi < I_B(v)$. The statement gives precisely the domain proof needed for $G(I_B(v))$.

Findings: none.

CMVBallCase.typeBPerimeterAtArea_strictMonoOn

Source hash: c3d6e28ec8a214fad8521a957798e0d4a463a65ea8d06f253bc35c5c42925728

Relation: equivalent

CMVBallCase.lean:1201-1219: The formal area domain is $(\lambda \cdot \pi, \infty)$. Area existence, strict area ordering of radii, and strict perimeter increase give the stated strict profile increase; no monotonicity of the difference of profiles is inferred.

Findings: none.

CMVBallCase.typeBPerimeterAtRadius

Source hash: c9c0993d1f5fbe4ba58debefa9858ea8f0b24ef4559a978342fbbc737aa8f2ff

Relation: equivalent

CMVBallCase.lean:321-325: The definition expands to the author's two weighted arc contributions with betaAtRadius and $\arccos(\lambda)$. The explanation correctly distinguishes this total expression from its geometric agreement theorem on $R \geq 0$.

Findings: none.

CMVBallCase.typeBPerimeterAtRadius_agrees

Source hash: fc3ebabe5ff5963057732d5c2e6cb24a284b9bf252ebb2ffea0fa55fa6ac0813

Relation: equivalent

CMVBallCase.lean:345-351: Both density bounds and $R \geq 0$ occur in Lean. Substitution of $\text{typeBRadius_betaAtRadius}$ identifies the radius perimeter with the angular perimeter, including the zero endpoint.

Findings: none.

CMVBallCase.typeBPerimeterAtRadius_continuous

Source hash: bb5bba6b7941dc1416656ad77a30a9cf4928feec80267c75bbd92466dad64edc

Relation: equivalent

CMVBallCase.lean:698-702: The formal theorem asserts global continuity for arbitrary real λ . With λ fixed, the chart and perimeter expression are continuous in R , including $R = 0$, as stated.

Findings: none.

CMVBallCase.typeBPerimeterAtRadius_deriv_pos

Source hash: 10de2cba5f52d44c61da385e12b1e3a912b33e8b23dd2e310115984cf7543ad2

Relation: equivalent

CMVBallCase.lean:677-684: Lean places $R > 0$ as an implication after the density assumptions; the author's equivalent quantified wording preserves it. The proof rewrites $\text{deriv}P_B$ to $2Q_B$ and uses factor positivity.

Findings: none.

CMVBallCase.typeBPerimeterAtRadius_hasDerivAt

Source hash: a36eb5dc07528a5ef5ac5c46d69519b24cf266609e428cac791ae feae3d159fc

Relation: equivalent

CMVBallCase.lean:441-464: Lean proves an ordinary derivative for all real R under $0 < \lambda < 1$. Expanding Q_B gives exactly its displayed derivative coefficient; the explanation correctly uses cancellation of the chart derivative terms.

Findings: none.

CMVBallCase.typeBPerimeterAtRadius_hasStrictDerivAt

Source hash: 89c1125258117d39f6582f2d8c1a2115de855d8212c426a1e171fed0a408f19b

Relation: equivalent

CMVBallCase.lean:2072-2089: Lean asserts HasStrictDerivAt with coefficient $2Q_B$ for arbitrary real R under admissible density. The explanation correctly defines strict differentiability as a two-point limit, not positivity, and derives it from the continuous ordinary derivative.

Findings: none.

CMVBallCase.typeBPerimeterAtRadius_strictMonoOn

Source hash: bcb0c4329c9e367c3237ca68300c9a926c8d4b286caa51d477c1d94e742ff2e8

Relation: equivalent

CMVBallCase.lean:731-739: Lean proves strict increase on $[0, \infty)$, using continuity at zero and interior derivative positivity. The author preserves the closed endpoint essential to the common-perimeter threshold.

Findings: none.

CMVBallCase.typeBPerimeterAtRadius_zero

Source hash: 4a8b3af699ffc34588703cbf0b3dfa0fe068d5c1e0ebc207f5e62ad1f86bcb22

Relation: equivalent

CMVBallCase.lean:759-766: The formal value is $2 \cdot \lambda \cdot \pi$. The endpoint chart and vanishing exterior radius leave the full weighted unit-circle boundary, exactly as explained.

Findings: none.

CMVBallCase.typeBPerimeterAt_typeBRadiusAtPerimeter

Source hash: 8c6523ab6857fea0ddfb73478b78aea70dd1ba779a838fdf73e197a8305abde2

Relation: equivalent

CMVBallCase.lean:1947-1951: The source equality is the stored type B perimeter equation at p. 1. Under the stated density and perimeter domain, this is exactly $P_B(r_B(p)) = p$, not an approximate inverse equation.

Findings: none.

CMVBallCase.typeBPerimeterValuesAtArea

Source hash: 7ffddd7593a4f7f71be09ace34b7c0443247b4c6e896e8cebb972d294f5834b0

Relation: equivalent

`CMVBallCase.lean`: 1068-1073: The definition existentially quantifies over every β with `BAdmissible`, area exactly v , and perimeter exactly p . The author's set notation preserves this quantifier order and makes no prior minimization or uniqueness assumption.

Findings: none.

CMVBallCase.typeBPerimeter_mem_values

Source hash: ab1cda17a1dd925edbcd6476abecb0e519a6fb9430ca150e790c745fb1bfa2ae

Relation: equivalent

`CMVBallCase.lean`: 1088-1092: The only hypotheses are angular admissibility and the area equation. The angle itself is the existential witness for its perimeter's membership; no density or uniqueness hypothesis is added.

Findings: none.

CMVBallCase.typeBRadius

Source hash: 495dd878203ed375c56ea0dbe5c390a5dfde6e2a9e8eec96e771b8ebff636634

Relation: equivalent

`CMVBallDefinitions.lean`: 14-17: The exact definition is $\sin(\text{typeBAlpha } \text{lam } \text{beta}) / \sin(\text{beta})$. Substituting α gives the author's formula; the common-chord relation and positive denominator on the admissible domain justify its geometric interpretation.

Findings: none.

CMVBallCase.typeBRadiusAtPerimeter

Source hash: 641ef25025d32c528f2e106a8f3ebc604806525be72db2a80847283bc6cc25ee

Relation: equivalent

`CMVBallCase.lean`: 1923-1930: The definition requires an `AdmissibleDensity` proof and a `CommonPerimeter` input and returns a subtype carrying positivity and the perimeter equation. It selects a witness only after uniqueness is proved; the author's uniquely characterized inverse is exact.

Findings: none.

CMVBallCase.typeBRadiusAtPerimeter_lt_typeCRadiusAtPerimeter

Source hash: 37fe08774ef3026b52e3131542e0b473bbc2fb7ecab7c6ff79923246a9265841

Relation: equivalent

`CMVBallCase.lean`: 1975-1991: The source proves strict order for every admissible density and genuine common perimeter. Assuming the reverse weak order contradicts $P_C(r_B) < P_B(r_B) = p$, exactly the author's argument.

Findings: none.

CMVBallCase.typeBRadiusAtPerimeter_pos

Source hash: 35f84c501e62b1b673fd6a9b793e1ea9f012ba0b179ed229e723b724fe0b5365

Relation: equivalent

CMVBallCase.lean:1937-1941: For admissible density and a genuine common perimeter, the formal conclusion extracts strict positivity from the inverse's codomain. The author's ordinary real notation faithfully suppresses only subtype packaging.

Findings: none.

CMVBallCase.typeBRadiusAtPerimeter_unique

Source hash: 45ce1f082ef5383bd43764a4e1518cd8d4004f8b048ca23100637518a3e785ee

Relation: equivalent

CMVBallCase.lean:1958-1965: The source additionally requires $R > 0$ and $P_B(R) = p \cdot 1$, and concludes the chosen inverse equals R . Existence-and-uniqueness supplies this equality in the stated orientation, used to identify local inverses.

Findings: none.

CMVBallCase.typeBRadius_betaAtRadius

Source hash: e66b10b3d091fb706b8fab34231c42e11270bceb8fc1f4044acdab9ecdf216f5

Relation: equivalent

CMVBallCase.lean:200-211: The source assumes admissible density and $R \geq 0$, then cancels the positive square root in the cotangent formula. The exact inverse identity and endpoint coverage are faithfully stated.

Findings: none.

CMVBallCase.typeBRadius_endpoint_zero

Source hash: 22146b3bf286e99a897eee49e1b29bb2805a205fd3eb8650ff9bf77fc6595ab6

Relation: equivalent

CMVBallCase.lean:278-283: There are no density assumptions in this Lean identity: α becomes π and `zero_div` evaluates the quotient to zero even at a totalized denominator. The author explicitly says all real λ and limits the geometric reading to admissible density.

Findings: none.

CMVBallCase.typeBRadius_eq_add_sqrt_mul_cot

Source hash: 3eb7e5731012db737e7b336d5d6d605bb445d1c2a4e8d3c0f094ba10820bb6da

Relation: equivalent

CMVBallCase.lean:81-91: The sine addition formula, $\cos(\arccos(\lambda)) = \lambda$, and positive $\sin(\beta)$ give $\lambda + \sqrt{1 - \lambda^2} \cdot \cot(\beta)$. Both density hypotheses and angular admissibility are present in the translation.

Findings: none.

CMVBallCase.typeBRadius_injective_on_admissible

Source hash: 0f30bd9be41461ee8fad9222c1b7b1550f99309779ba29cd8036dd728bb91efb

Relation: equivalent

CMVBallCase.lean:120-126: Both angles must be admissible and the density lies in $(0, 1)$. Strict decrease yields equality of angles from equality of radii, matching the statement used to reconstruct candidates.

Findings: none.

CMVBallCase.typeBRadius_nonneg

Source hash: 3582dff45831f2e7f5c8a52df6dd4d3c3ab5060e81b548fb6a841a2254794a31

Relation: equivalent

CMVBallCase.lean:232-240: The proof divides the nonnegative sine of α by the positive sine of β . Lean and the author both conclude nonnegativity, preserving the possible zero-radius endpoint.

Findings: none.

CMVBallCase.typeBRadius_strictAntiOn

Source hash: d474e75d38db799e62e6a1fd3b84d82959bcc51226ab32f06c71df9e248314af

Relation: equivalent

CMVBallCase.lean:103-119: The formal set is all BAdmissible angles, including $\beta = \pi - \arccos(\lambda)$. The proof combines the global cotangent identity with a positive square-root coefficient; the author's whole-domain strict decrease is exact.

Findings: none.

CMVBallCase.typeB_perimeter_factor_pos

Source hash: ac194654106ebe6db799026b01d031f3446dcc24e3fc1b5dfd81512dcf10bc66

Relation: equivalent

CMVBallCase.lean:623-643: Under admissible density and $R > 0$, the complementary angle $\pi - \beta$ lies strictly in $(0, \pi)$. Its segment area is exactly Q_B , giving the strict sign used in derivatives.

Findings: none.

CMVBallCase.typeB_sin_pos

Source hash: 778d4c7a3220c8e895cdfc64fd5909fe596a22442647d2cfb4c145e12e3651e

Relation: equivalent

CMVBallCase.lean:63-71: Admissibility together with $\arccos(\lambda) > 0$ places β strictly between 0 and π . The stated density and admissibility hypotheses match Lean and justify nonzero radius denominators.

Findings: none.

CMVBallCase.typeB_values_eq_singleton

Source hash: 0aaa46dd0b8b451605797b9ef606b1e37cf80fd960f4abcc25d735f41934d431

Relation: equivalent

CMVBallCase.lean:1098-1110: Lean assumes an admissible β of area v and a universal implication identifying every other admissible angle of that area with β . The author's uniqueness wording and conditional set equality preserve all these hypotheses.

Findings: none.

CMVBallCase.typeCArea

Source hash: 1c838263bf547723f2086f0226358db55c1fd829ed4addfba436f74cda3734ab

Relation: equivalent

CMVBallDefinitions.lean:37-43: The source quotient and final subtracted term agree exactly with the author. The identity $S(\pi/2 - \beta) = \pi/2 - \beta - \sin(\beta)\cos(\beta)$ confirms the lens correction and its coefficient $(1 - \lambda)$, with no factor-of-two error.

Findings: none.

CMVBallCase.typeCAreaAtRadius

Source hash: 348b519a27228df7e22a823dc6b7c50b1974e565424be0185003a97e07856f1d

Relation: equivalent

CMVBallCase.lean:339-344: $S(\pi/2 - b) = \pi/2 - b - \sin(b)\cos(b)$, so the author's segment notation is algebraically identical to Lean's last term. The overlap correction and both coefficients are exact.

Findings: none.

CMVBallCase.typeCAreaAtRadius_agrees

Source hash: ef9487e571bf7a8e9ecfdce2224335bbe39d0b9603b9c566eb6c213c2b58a4f8

Relation: equivalent

CMVBallCase.lean:367-377: For $R > 0$ and arbitrary λ , the proof converts the inverse square of tangent into R^2 . The displayed equality and lack of a density restriction agree with Lean.

Findings: none.

CMVBallCase.typeCAreaAtRadius_continuousOn

Source hash: c92a0518d3bc80f606df13b4b6f5c21e7bd9c2747515397578edebe4a4433736

Relation: equivalent

CMVBallCase.lean:703-707: The source permits arbitrary λ and asserts continuity only on $\text{Ioi } 0$. It derives this from positive-radius differentiability; the author does not incorrectly extend it across the totalized zero input.

Findings: none.

CMVBallCase.typeCAreaAtRadius_deriv_pos

Source hash: 5b0c9033c9a26a0f21e15311d7c9b43d1a45758e4340fc63a1e1040bf167f9d8

Relation: equivalent

CMVBallCase.lean:670-676: Admissible density and positive R make the established derivative $2RQ_C$ strictly positive. The translation has exactly these hypotheses and the same ordinary derivative conclusion.

Findings: none.

CMVBallCase.typeCAreaAtRadius_hasDerivAt

Source hash: b473b5e3212cdaca79ca31990e26c74cf5808e134b828057b642784ffa2747a8

Relation: equivalent

CMVBallCase.lean:572-609: Lean requires only $R > 0$ and permits all real λ . The complementary-angle identity and $\cos(b) = R\sin(b)$ cancel the overlap derivatives, yielding $R \cdot (2Q_C)$, as translated.

Findings: none.

CMVBallCase.typeCAreaAtRadius_large_lower

Source hash: b45b5794cbb45181f9541a2d6aaedfb79df716a6e045c02aa69ba8995a958442

Relation: equivalent

CMVBallCase.lean:833-865: The coefficient of R^2 is greater than $\lambda \cdot \pi$ and the last deficit is less than $\pi/2$. The stated strict quadratic lower bound, density assumptions, and positive-radius domain all match Lean.

Findings: none.

CMVBallCase.typeCAreaAtRadius_lt_full

Source hash: a7ac56a00483dd17c810d49f7f3b39998e96e24cbcc05612f96aba66e39e49b6

Relation: equivalent

CMVBallCase.lean:810-832: The formal result is $A_C(R) < \pi \cdot R^2$ for admissible density and $R > 0$. Both overlap segments and their density deficit are positive, exactly the geometric/algebraic explanation given.

Findings: none.

CMVBallCase.typeCAreaAtRadius_strictMonoOn

Source hash: bbc5de3a1bc43d4360f2a79a6c274beccf88985cbfcb7b993a4fe30a532f86a2

Relation: equivalent

CMVBallCase.lean:722-730: The source domain is the open positive half-line. The positive derivative and continuity prove strict area increase between any two positive radii, with no assertion at zero.

Findings: none.

CMVBallCase.typeCPerimeter

Source hash: dae34722c1182356b6a588c3b1c13561ad7e927882e5ef9397cd8e01e6fe4454

Relation: equivalent

CMVBallDefinitions.lean:33-36: The formal expression $2 \cdot (\pi - (1 - \lambda) \cdot \beta) / \tan(\beta)$ agrees exactly with the translation. The inner-arc deficit from a circle of radius $\cot(\beta)$ gives the same weighted perimeter as CMV p. 35.

Findings: none.

CMVBallCase.typeCPerimeterAtAArea

Source hash: 2e3dfe31281e09c4e4e68ab71c76cfa596f35e20fe535735ebbe9fcbb7c59fc8

Relation: equivalent

CMVBallCase.lean:1085-1087: The formal profile is exactly $s\text{Inf}$ of the type C value set. The author does not replace this with an unexplained selected radius and correctly states why totalization is irrelevant where the theorem applies.

Findings: none.

CMVBallCase.typeCPerimeterAtAArea_eq

Source hash: 910202180ff5ab8baae5e253008da92ce732b62fac764c99dfe1e8b67dffc4a3

Relation: equivalent

CMVBallCase.lean:1131-1137: Lean derives the actual candidate perimeter from the singleton value set under the same three hypotheses. The author's statement neither assumes unconditional uniqueness nor loses the exact area constraint.

Findings: none.

CMVBallCase.typeCPerimeterAtAArea_eq_radius

Source hash: 67e8363dd82abecb62d39098aa381615d4943a0c48b8f9f5ecc2be957cbd9e7a

Relation: equivalent

CMVBallCase.lean:1183-1200: The formal assumptions are admissible density, $R > 0$, and $A_C(R) = v$. All other admissible angles are shown to have the same positive radius and angle; the profile evaluation is exact without an extra lower bound on v .

Findings: none.

CMVBallCase.typeCPerimeterAtAArea_lt_of_commonPerimeterAreaGap_neg

Source hash: 6ed9af03cd6961ea327cfb3f5237fc3e86432b34c2cd4ca2282daeb7016ccc0a

Relation: equivalent

CMVBallCase.lean:2378-2422: The formal strict premise $G(I_B(v)) < 0$ means A_C at that common perimeter exceeds v . Strict area and perimeter monotonicity give $I_C(v) < I_B(v)$; all density and area-domain hypotheses are preserved.

Findings: none.

CMVBallCase.typeCPerimeterAtRadius

Source hash: 5bfe6bbb6f40601afee63cea3bd361fb5c2ec4e0cae513f610cfd4255fe4ae2f

Relation: equivalent

CMVBallCase.lean:335-338: The definition is exactly $2 * R * (\pi - (1 - \lambda_m) * b\text{AtRadius}(R))$. The explanation correctly obtains it from the full circumference by subtracting the density deficit of the inner arc.

Findings: none.

CMVBallCase.typeCPerimeterAtRadius_agrees

Source hash: 07f1596dce2184a5c103c4a465e9a7f508c9fbe84d173c9f12c79dc46e54a1a2

Relation: equivalent

CMVBallCase.lean: 358-366: Lean assumes only $R > 0$ and allows arbitrary real λ . The explanation preserves that domain and uses the reciprocal tangent identity to obtain exact perimeter equality.

Findings: none.

CMVBallCase.typeCPerimeterAtRadius_continuousOn

Source hash: 8ba982df91d14a4376354beb85fd26540d8e9d0d83b34015727e5932cb96b536

Relation: equivalent

CMVBallCase.lean: 708-712: For every real λ the formal domain is $(0, \infty)$. The proof uses ordinary differentiability at positive radii; the translation preserves precisely that asserted domain.

Findings: none.

CMVBallCase.typeCPerimeterAtRadius_deriv_pos

Source hash: 0a0f4d788d01de74c363719e0d086cb74d4be34b638c450f00b59a31d9f27b99

Relation: equivalent

CMVBallCase.lean: 685-692: The formal implication assumes positive radius in addition to admissible density. The proof obtains $\text{deriv}P_C = 2Q_C > 0$, exactly the nonvanishing used for inversion.

Findings: none.

CMVBallCase.typeCPerimeterAtRadius_gt_pi_mul_radius

Source hash: 096d139a89edea080509b072e5206625c2521a1d0213be4b9a082a2bd82369f3

Relation: equivalent

CMVBallCase.lean: 1841-1852: The source assumes $0 < \lambda < 1$ and $R > 0$. Its coefficient bound $\pi/2 < \pi - (1 - \lambda)b(R)$ gives $\pi \cdot R < P_C(R)$, preserving the strict lower bound and unbounded-growth use in the explanation.

Findings: none.

CMVBallCase.typeCPerimeterAtRadius_hasDerivAt

Source hash: f5d2c7b26d784940465bfecd68ce738beed977fd9837beb85c6df684f2f89412

Relation: equivalent

CMVBallCase.lean: 465-487: The source assumes $R > 0$ but no density bounds. Its coefficient is exactly $2Q_C$, and the author separates this derivative identity from the later positivity theorem.

Findings: none.

CMVBallCase.typeCPerimeterAtRadius_hasStrictDerivAt

Source hash: 3e2b1b6740838724efc3c00deba8f2d16532a3855f0a5cf96b2667b1a8855fae

Relation: equivalent

CMVBallCase.lean: 2090-2107: Both density bounds and $R > 0$ appear in this source theorem. Nearby positive-radius derivatives have continuous coefficient $2Q_C$, yielding the stated strict two-point derivative needed for local inversion.

Findings: none.

CMVBallCase.typeCPerimeterAtRadius_strictMonoOn

Source hash: ebda1658a94a124439fef17a703088753a165f7a75d443b97607309192ac577f

Relation: equivalent

CMVBallCase.lean: 740-748: The formal statement is `StrictMonoOn` on $(0, \infty)$, with both density bounds. The explanation correctly transports strict order using the mean value theorem on positive-radius intervals.

Findings: none.

CMVBallCase.typeCPerimeterAt_typeCRadiusAtPerimeter

Source hash: 97f0a09e977f148c1547234ee78701dc58ab72d702573d3d019190ba9b227ce1

Relation: equivalent

CMVBallCase.lean: 1952-1957: The formal equality projects the perimeter component of the chosen type `C` radius's characterization. The author's equation and admissible input domain match exactly.

Findings: none.

CMVBallCase.typeCPerimeterValuesAtArea

Source hash: cb485477b22802c53c6af448371d844cdb35a1ad16956ed8dcc86d1d34b9f786

Relation: equivalent

CMVBallCase.lean: 1074-1079: The formal set ranges over every `CAdmissible` angle of area v and collects its actual perimeter. The explanation correctly includes the full open angular domain and no selected inverse branch.

Findings: none.

CMVBallCase.typeCPerimeter_mem_values

Source hash: ad514705a9dc71bb1959ff8334149fb5759b7a8f9d6bb4f82c2b59f07695ab1c

Relation: equivalent

CMVBallCase.lean: 1093-1097: The source requires `CAdmissible` and the area equation and concludes membership of that same angle's perimeter. The direct existential witness and unrestricted density scope are faithfully explained.

Findings: none.

CMVBallCase.typeCRadius

Source hash: a418ec9c9727a88b5945a9e22b827624dc661a72d7ca8532d18c95ff79188dd8

Relation: equivalent

CMVBallDefinitions.lean:30-32: The definition is exactly $1/\tan(\beta)$. The author restricts the orthogonal-ball interpretation to $0 < \beta < \pi/2$ and correctly avoids interpreting Lean's totalized division at either excluded angular endpoint.

Findings: none.

CMVBallCase.typeCRadiusAtPerimeter

Source hash: a58a803701a0a2d5cf3a9913ddb7c5b645d75bedddb7e49e42f7c5d55ebc49a7

Relation: equivalent

CMVBallCase.lean:1931-1936: The source selects the unique positive-radius solution on the common-perimeter subtype and stores its defining equation. The explanation correctly excludes values at or below $2\lambda \cdot \pi$ and does not imply geometric branch choice.

Findings: none.

CMVBallCase.typeCRadiusAtPerimeter_pos

Source hash: 6d7f4a31d01f4556bf0e8a9719b79bf61ccb4965c124289a537df2d839c9fd39

Relation: equivalent

CMVBallCase.lean:1942-1946: The inverse's stored characterization gives a strictly positive real radius for every admissible input. The author preserves both the density and open-perimeter hypotheses.

Findings: none.

CMVBallCase.typeCRadiusAtPerimeter_unique

Source hash: 6d0bba8a25476b26089b665ae685627f40cb11720cbb717de9b249a9d0e5a11c

Relation: equivalent

CMVBallCase.lean:1966-1974: The formal candidate is positive and solves the same perimeter equation as the subtype inverse. Uniqueness gives $r_c(p) = R$ under precisely the translated hypotheses.

Findings: none.

CMVBallCase.typeCRadius_bAtRadius

Source hash: 6e7442e12b01ef5ceace4300d1d77efe39044855122de67c6d01f583bea0c6a8

Relation: equivalent

CMVBallCase.lean:290-295: With $R > 0$, $\tan(\arctan(1/R)) = 1/R$ and reciprocal cancellation prove radius recovery. The author preserves the positive-radius hypothesis even though the underlying total algebra can be broader.

Findings: none.

CMVBallCase.typeCRadius_injective_on_admissible

Source hash: c8f101f2d5546c0e443cf5cbb59e42ff32eecef077bdf0c84c0da4896e2b3865

Relation: equivalent

CMVBallCase.lean:48-52: Set.InjOn is injectivity of typeCRadius on all CAdmissible angles, not an existential choice of branch. The proof obtains it from strict decrease as described.

Findings: none.

CMVBallCase.typeCRadius_pos

Source hash: 1a7173c1755709ae7faf95a9a6109b9e8aa56a4449eb1eeb91c7af81f092911c

Relation: equivalent

CMVBallCase.lean:28-35: The sole hypothesis is CAdmissible beta. The proof takes the reciprocal of the positive tangent on $(0, \pi/2)$; the informal statement adds no density condition.

Findings: none.

CMVBallCase.typeCRadius_strictAntiOn

Source hash: 5aff3dac04279f6fbf34d1285134552fc7dd876dc06d4b5558e36e5ba717c3c1

Relation: equivalent

CMVBallCase.lean:36-47: StrictAntiOn quantifies over every ordered pair in $(0, \pi/2)$. The proof uses strict increase and positivity of tangent, giving the stated strict reciprocal inequality throughout that interval.

Findings: none.

CMVBallCase.typeC_dominance_persists

Source hash: f5ada2cd0133c47ab6c9f8734b2bc2a998e82cc392fe14d449f57858a255b00e

Relation: equivalent

CMVBallCase.lean:2423-2448: The universally quantified real λ , v_0 , v and all five hypotheses match the frozen root exactly: $0 < \lambda < 1$, $\lambda \cdot \pi < v_0 < v$, and weak initial dominance. The proof obtains $I_B(v_0) < I_B(v)$, then a strictly negative later common-perimeter gap and strict later dominance, without assuming a crossing or geometric classification.

Findings: none.

CMVBallCase.typeC_perimeter_factor_pos

Source hash: c725d934df891e67e5699ac456ab0db215e125103ab9c675b3e231a6c66930e3

Relation: equivalent

CMVBallCase.lean:644-662: The formal coefficient is Q_C . Acute-angle sine/cosine positivity bounds $S(b)$ between zero and $b < \pi/2$, and $0 < 1 - \lambda < 1$ proves strict positivity under the stated hypotheses.

Findings: none.

CMVBallCase.typeC_values_eq_singleton

Source hash: afcbdbc6bce67fe83c6357fe4f5677ffc4fb903b985aa4822c1eef450d8ed5ce

Relation: equivalent

CMVBallCase.lean:1111-1123: The formal universal uniqueness condition ranges over all admissible angles of area v . Along with the given candidate, it proves exactly the stated singleton equality, not independent existence or uniqueness.

Findings: none.

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